

Quantum Algorithms Summary of Recent Progress on QEC

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In general, the Google paper[1] showed current hardware can run surface code and do error correction, but surface code is not good at scaling. The IBM paper[2] show a protocol that scales well but cannot be implemented on presently available hardware. The Quantinuum paper[3] showed a way to do error corrected computation in ion-trap quantum computer but did not show universal quantum operation. The Microsoft paper[4] show error corrected quantum computing with neutral atom, entanglement and connectivity is good but error rate and cycle number is not good.

Quantum Error Correction Below the Surface Code Threshold – the Google paper

Key takeaway

This paper reports the experimental demonstration of surface code error correction with below-threshold physical qubits. As well as experiments in repetition code that demonstrates improvement in performance.

A surface code memory below threshold

Distance-3, distance-5, and distance-7 surface codes are demonstrated.

Logical performance is characterized by fitting the logical error per cycle ϵ_d (up to 250 cycles), averaged over logical eigenstates of X_L and Z_L bases. For $d = 3, 5$, results from distance-3 and distance-5 subgrids are compared to the distance-7 code.

Λ is computed via linear regression of $\ln[\epsilon_d]$, where ϵ_d is the logical error per cycle. With a neural network decoder, the error suppression factor is around 2.

Distance-7 logical qubit lifetime surpasses constituent physical qubit lifetimes, extending prior results from bosonic to multiqubit codes. Significantly advances towards logical operation breakeven.

A repetition code memory in the ultralow-error regime

Running repetition code showed logical error floors around 10^{-10} . But repetition code does not correct enough types of error simultaneously.

Coherent error injection confirmed logical error scaling consistent with $O(p^{\frac{d+1}{2}})$ suppression.

Real-time decoding

The logical operations are non-deterministic, logical measurement outcomes must be correctly interpreted on the fly, therefore classical co-processor that can decode errors in real time is essential

to fault-tolerant quantum computing.

Their decoder latency on average is $63\mu s$

Outlook

This paper report experiment on new quantum processors that shows it is possible to execute fault-tolerant quantum computing by employing surface code. It shows good progress are being made in the execution performance of quantum processor.

Logical Computation Demonstrated with a Neutral Atom Quantum Processor – the Microsoft paper

Key takeaway

They did experiments with a neutral atom quantum processor with 256 qubits. They implemented logical qubits and showed it has a smaller error rate than baselined unencoded qubits. They entangled 24 logical qubits in a "cat" state, which is the largest to date. They implement the Bernstein-Vazirani algorithm for oracle up to 28 logical qubits encoded in 112 physical qubits.(showing "improved efficiency compared to classical algorithms") They showed fault-tolerance and error correction, using distance-three Bacon-Shor code.

Logical Error Rate and Error Correction

4.9% after one error correction cycle, 8% after 2. 0.5 errors is corrected per round. 0.88 is corrected in two rounds.

To compare, teleportation of logical qubit is done, it is compared with the results for repeated teleportation of physical qubit states.

Bernstein-Vazirani Algo. is done

Bernstein-Vazirani algorithm experiments used encoded logical qubits (based on J4,1,2K code) and unencoded circuits. Results showed encoded computation outperformed unencoded circuits under noise and loss conditions, improving accuracy for determining the secret string ($s = 1^n$).

Outlook

1. Highlighted loss-to-erasure conversion effectiveness, indicating that future neutral atom quantum computers could efficiently operate even with significant atom loss.
2. Foresees advances in two-qubit gate fidelities by reducing gate laser noise and expects substantial scale-up, approaching 10,000 qubits. This will enable higher-distance codes, more robust logical qubits, and improved fault-tolerant universal quantum computing demonstrations.

Demonstration of quantum computation and error correction with a tesseract code – the Quantinuum paper

Key takeaway

In this paper they introduce a "tesseract subsystem color code". They talk about the innovation, experiments and results, advantages and limitations.

Tesseract subsystem color code

Tesseract subsystem color code is a quantum error-correcting code that encodes four logical qubits into 16 physical qubits at a distance-four fault tolerance. In this paper their innovation is simplifying error correction by sacrificing two logical qubits (acting as gauge qubits), reducing stabilizer measurements from weight-eight to weight-four, enhancing error correction efficiency and enabling simultaneous correction of X and Z errors.

Experiments

Experiments demonstrated substantial logical error reduction:

- 4 qubits: Targeted encoded CNOT gate within a single block; 15-fold improvement.
- 8 qubits: Deeper circuits with transversal CNOT gates; 11-fold improvement.
- 12 qubits: Entangled logical GHZ state; 22-fold improvement.

Additionally, multi-round error correction experiments achieved up to 24-fold improvement for 4 qubits and 8-fold for 8 qubits, validating the effectiveness of the tesseract subsystem color code in fault-tolerant quantum computing.

Advantage and limitation

The tesseract subsystem color code significantly reduces logical errors compared to unencoded circuits. Simplified measurements enable efficient single-shot, fault-tolerant error correction and direct logical operations within single code blocks, avoiding teleportation overhead.

Due to distance-four coding, postselection rejects multiple-fault trials, slightly lowering acceptance rates and complicating scaling. Universal quantum operations and inter-block operations remain experimentally untested.

High-threshold and Low-overhead Fault-tolerant Quantum Memory – the IBM paper

Key Takeaway

They made a new quantum error correction protocol based on Low-Density Parity-Check (LDPC) codes, specifically a new family called Bivariate Bicycle (BB) codes. It is very qubit-efficient.

BB Code Structure and Syndrome Measurement

BB codes use a degree-6 Tanner graph with thickness-2, enabling efficient connectivity for hardware implementation. Syndrome measurements are performed with depth-7 circuits using layers of CNOT gates to couple data and check qubits.

Connectivity

Each physical qubit is coupled to exactly six others, forming a connectivity described by a "thickness-2" graph, decomposed into two planar subgraphs. This connectivity is practical for superconducting qubit architectures.

Threshold

The BB code described achieves a threshold error rate of around 0.7%

Decoder

The proposed protocol uses belief propagation with ordered statistics decoder (BP-OSD). The decoder first precomputes expected syndromes for single errors. Then Uses actual measured syndromes during quantum operations to quickly guess and correct errors.

Efficiency

The paper demonstrates significant efficiency gains: for example, encoding 12 logical qubits with the BB code $[[144, 12, 12]]$ requires 288 physical qubits, a dramatic improvement over the nearly 3000 qubits required by traditional surface codes for similar performance.

Integration with surface code

The BB codes support logical operations (e.g., logical X and Z measurements) by extending their Tanner graph, integrating them with surface codes for load-store operations via quantum teleportation

Let's not get too excited just yet

It remains unclear whether our code examples can be scaled up while retaining the high encoding rate in the limit of large code length.

Outlook

This paper proposes BB LDPC quantum codes offering practical fault-tolerant quantum memory with significantly lower qubit overhead compared to surface codes, enabling near-term quantum computing demonstrations. However, the implementation requires overcoming technical challenges such as developing multi-layered superconducting chips, qubits supporting higher connectivity, and engineering efficient, long-range microwave couplers for superconducting architectures.

References

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- [3] Reichardt, Ben W., et al. "Demonstration of quantum computation and error correction with a tesseract code." arXiv preprint arXiv:2409.04628 (2024).
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